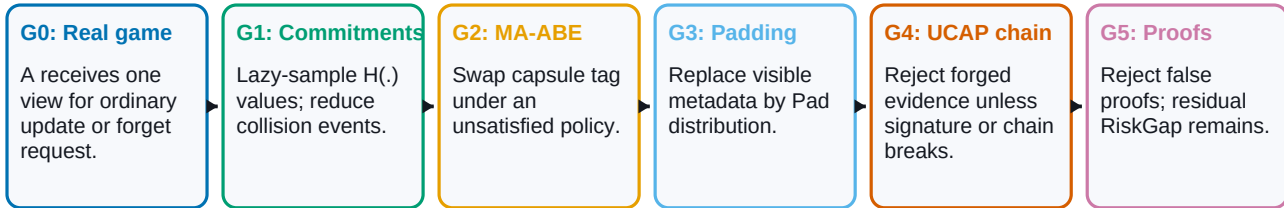


Policy-authenticated update-hiding game and reduction purpose

Each hop bounds one observable advantage: commitments, policy capsule, channel shape, audit forgery, and false proofs.



Reduction bound

$$\text{Adv}_{\text{auth}}(A) \leq \text{Adv}_{\text{MA-ABE}}(B1) + \text{Adv}_{\text{RO}}(B2) + \text{Adv}_{\text{PAD}}(B3) + \text{Adv}_{\text{SIG}}(B4) \\ + \text{Adv}_{\text{ZK}}(B5) + \epsilon_R + \text{negl}(\lambda)$$

B1: policy capsule indistinguishability

B2: commitment consistency

B3: request-type padding

B4: audit-chain unforgeability

B5: authorization/repair proof soundness